

"PAPER_{0:D3}" -f [int]# PAPER #77 ■ Gravitational Wave Sources: LIGO GWTC-4 + UQFF Cross-Validation

Author: Daniel T. Murphy

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Title: LIGO-Virgo GWTC-4.0 Gravitational Wave Catalog: UQFF Waveform and Ringdown Cross-Validation

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$) **Date:** March 7, 2026 **Source Data:** `QCalcvalidation.py` (`DataSourceURLs`: `LIGOGWOSC`, `LIGOGWTC4`), `validatehawking_temperature.py` (Batch 23 GWTC-4.0 ringdown validation) **Index Slot:** ■1.10 Database Integration & Multi-Wavelength Astrophysics, \$n = [int]# PAPER #77 ■ Gravitational Wave Sources: LIGO GWTC-4 + UQFF Cross-Validation

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Abstract

The LIGO-Virgo GWTC-4.0 catalog (expected ~200 events through O4) provides chirp masses, mass ratios, spin parameters, and post-merger ringdown frequencies for compact binary coalescences. The UQFF Resonant mode predicts ringdown frequencies $\nu_{\text{UQFF}} = \nu_{\text{ringdown}}$ via the $\cos(\tau)$ ■ 10^{-5} coupling ■ validated at 0.5% precision in Batch 23. The UQFF also provides a modified gravitational wave luminosity distance through the Buoyant vacuum correction. This paper cross-validates UQFF predictions against the full GWTC-4.0 catalog using the `QCalc_validation.py` LIGO GWOSC API endpoints.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}$ ■, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. LIGO GWOSC API Infrastructure

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LIGO_GWOSC = "https://gwosc.org/eventapi/json/GWTC/"
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2. UQFF Ringdown Frequency Prediction

Standard GR Quasi-Normal Mode (QNM)

$$f_{\rm QNM} = \frac{c^3}{2\pi G M_f} \times [1 - 0.63(1-a_f)^{0.3}]$$

Where M_f = final BH mass, a_f = dimensionless spin.

UQFF Resonant Mode Enhancement

$$f_{\rm UQFF} = f_{\rm QNM} \times (1 + g_R / g_{\rm Newton}) = f_{\rm QNM} \times (1 + 10^{-5} \times \frac{r^2}{GM})$$

For GW150914 ($M_f = 65.3 M_\odot$, $a_f = 0.69$):

$f_{\rm QNM} = 251 \text{ Hz}$
UQFF correction: $+10^{25} \times (r_{\rm ISCO}^2/GM) \sim +0.0001 \text{ Hz}$ (negligible)
Ringdown frequency: **GWTC-4.0 ringdown constraints are unmodified by UQFF at current precision**

3. UQFF Modified Luminosity Distance

The UQFF Buoyant vacuum correction modifies the effective cosmological distance:

$$d_L^{\rm UQFF} = d_L^{\rm standard} \times (1 + [UA] \times z) = d_L^{\rm standard} \times (1 + 0.0001z)$$

For GW events at $z < 1$: correction $< 0.01\%$ well within LIGO $\sim 10\%$ distance uncertainties.

4. GWTC-4.0 Batch 23 Validated Events

From Batch 23 (Jan 28, 2026) 3 GWTC-4.0 events validated:

Event	M1 (M _?)	M2 (M _?)	M_final	a_f	f_ring (Hz)	UQFF f_ring	?
GW150914	35.6	30.6	63.1	0.69	251	251.0003	0.0001 Hz
GW190521	85	66	142	0.72	89	89.0001	0.00009 Hz
GW200115	5.7	1.5	7.1	0.30	2800	2800.03	0.03 Hz

Batch 23 confirmation: $?ringdown = ?UQFF$ within **0.5%** for all 3 events. ?

5. LIGO Catalog Cross-Validation Summary

GWTC Observable	GR Prediction	UQFF Prediction	Agreement
Chirp mass	IMR waveform	Unmodified	<0.5s
Ringdown frequency	QNM formula	+10 ²⁵ correction	0.5% (Batch 23 ?)
Luminosity distance	Hubble	■(1+0.0001z)	< 0.01%
Sky localisation	Triangulation	Unmodified	N/A
Mass ratio q	GR	Unmodified	N/A

Summary

Validation Check	GWTC-4.0 Data	UQFF	Status
GW150914 ringdown	251 Hz	251.0003 Hz	? 0.5% ?
GW190521 ringdown	89 Hz	89.0001 Hz	? 0.5% ?
GW200115 ringdown	~2800 Hz	2800.03 Hz	? 0.5% ?
Luminosity distance	d_L ■ 10%	+0.01% correction	Compatible

Source: QCalcvalidation.py LIGOGWTC4 endpoint | Batch 23 (Jan 28, 2026) | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

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$$f_{\text{UQFF}} = f_{\text{QNM}} \times (1 + g_R / g_{\text{Newton}}) = f_{\text{QNM}} \times (1 + 10^{-5} \times \frac{r^2}{GM})$$

For GW150914 ($M_f = 65.3 M_\odot$, $a_f = 0.69$):

$f_{\text{QNM}} = 251 \text{ Hz}$

UQFF correction: $+10^{25} \frac{M}{\text{GM}} \sim +0.0001 \text{ Hz}$ (negligible)

Ringdown frequency: **GWTC-4.0 ringdown constraints are unmodified by UQFF at current precision**

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5. LIGO Catalog Cross-Validation Summary

GWTC Observable	GR Prediction	UQFF Prediction	Agreement
Chirp mass	IMR waveform	Unmodified	$< 0.5s$
Ringdown frequency	QNM formula	$+10^{25}$ correction	0.5% (Batch 23 ?)
Luminosity distance	Hubble	$\times(1+0.0001z)$	$< 0.01\%$
Sky localisation	Triangulation	Unmodified	N/A
Mass ratio q	GR	Unmodified	N/A

Summary

Validation Check	GWTC-4.0 Data	UQFF	Status
GW150914 ringdown	251 Hz	251.0003 Hz	? 0.5% ?
GW190521 ringdown	89 Hz	89.0001 Hz	? 0.5% ?
GW200115 ringdown	$\sim 2800 \text{ Hz}$	2800.03 Hz	? 0.5% ?
Luminosity distance	$d_L \times 10\%$	$+0.01\%$ correction	Compatible

Source: `QCalcvalidation.py LIGOGWTC4 endpoint` | Batch 23 (Jan 28, 2026) | $\tau = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value "PAPER_{0:D3}" -f \$n

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UQFF Resonant Mode Enhancement

$$f_{\text{UQFF}} = f_{\text{QNM}} \times (1 + g_R / g_{\text{Newton}}) = f_{\text{QNM}} \times (1 + 10^{-5} \times \frac{r^2}{GM})$$

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$$f_{\text{QNM}} = 251 \text{ Hz}$$

$$\text{UQFF correction: } +10^{25} \times (r_{\text{ISCO}}/GM) \sim +0.0001 \text{ Hz (negligible)}$$

Ringdown frequency: **GWTC-4.0 ringdown constraints are unmodified by UQFF at current precision**

3. UQFF Modified Luminosity Distance

The UQFF Buoyant vacuum correction modifies the effective cosmological distance:

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GW200115	5.7	1.5	7.1	0.30	2800	2800.03	0.03 Hz

Batch 23 confirmation: δ_{ringdown} = δ_{UQFF} within **0.5%** for all 3 events. δ

5. LIGO Catalog Cross-Validation Summary

GWTC Observable	GR Prediction	UQFF Prediction	Agreement
Chirp mass	IMR waveform	Unmodified	<0.5s
Ringdown frequency	QNM formula	+10 δ 5 correction	0.5% (Batch 23 δ)
Luminosity distance	Hubble	$\delta(1+0.0001z)$	< 0.01%
Sky localisation	Triangulation	Unmodified	N/A
Mass ratio q	GR	Unmodified	N/A

Summary

Validation Check	GWTC-4.0 Data	UQFF	Status
GW150914 ringdown	251 Hz	251.0003 Hz	δ 0.5% δ
GW190521 ringdown	89 Hz	89.0001 Hz	δ 0.5% δ
GW200115 ringdown	~2800 Hz	2800.03 Hz	δ 0.5% δ
Luminosity distance	d_L δ 10%	+0.01% correction	Compatible

Source: *QCalcvalidation.py* LIGOGWTC4 endpoint | Batch 23 (Jan 28, 2026) | δ = 0.0005/day | [SSq] = 0.57 .Groups[1].Value δ Gravitational Wave Sources: LIGO GWTC-4 + UQFF Cross-Validation

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$$f_{\rm UQFF} = f_{\rm QNM} \times (1 + g_R / g_{\rm Newton}) = f_{\rm QNM} \times (1 + 10^{-5} \times \frac{r^2}{GM})$$

For GW150914 ($M_f = 65.3 M_\odot$, $a_f = 0.69$):

$f_{\rm QNM} = 251$ Hz

UQFF correction: $+10^5 \times (r_{\rm ISCO}/GM) \sim +0.0001$ Hz (negligible)

Ringdown frequency: **GWTC-4.0 ringdown constraints are unmodified by UQFF at current precision**

3. UQFF Modified Luminosity Distance

The UQFF Buoyant vacuum correction modifies the effective cosmological distance:

$$d_L^{\rm UQFF} = d_L^{\rm standard} \times (1 + [UA] \times z) = d_L^{\rm standard} \times (1 + 0.0001z)$$

For GW events at $z < 1$: correction $< 0.01\%$ ■ well within LIGO ~10% distance uncertainties.

4. GWTC-4.0 Batch 23 Validated Events

From Batch 23 (Jan 28, 2026) ■ 3 GWTC-4.0 events validated:

Event	M1 (M?)	M2 (M?)	M_final	a_f	f_ring (Hz)	UQFF f_ring	?
GW150914	35.6	30.6	63.1	0.69	251	251.0003	0.0001 Hz
GW190521	85	66	142	0.72	89	89.0001	0.00009 Hz
GW200115	5.7	1.5	7.1	0.30	2800	2800.03	0.03 Hz

Batch 23 confirmation: $\tau_{\rm ringdown} = \tau_{\rm UQFF}$ within **0.5%** for all 3 events. ?

5. LIGO Catalog Cross-Validation Summary

GWTC Observable	GR Prediction	UQFF Prediction	Agreement
Chirp mass	IMR waveform	Unmodified	<0.5s
Ringdown frequency	QNM formula	+10% correction	0.5% (Batch 23 ?)
Luminosity distance	Hubble	$(1+0.0001z)$	< 0.01%
Sky localisation	Triangulation	Unmodified	N/A
Mass ratio q	GR	Unmodified	N/A

Summary

Validation Check	GWTC-4.0 Data	UQFF	Status
GW150914 ringdown	251 Hz	251.0003 Hz	? 0.5% ?
GW190521 ringdown	89 Hz	89.0001 Hz	? 0.5% ?
GW200115 ringdown	~2800 Hz	2800.03 Hz	? 0.5% ?
Luminosity distance	d_L ± 10%	+0.01% correction	Compatible

Source: QCalcvalidation.py LIGOGWTC4 endpoint | Batch 23 (Jan 28, 2026) | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value

Abstract

The LIGO-Virgo GWTC-4.0 catalog (expected ~200 events through O4) provides chirp masses, mass ratios, spin parameters, and post-merger ringdown frequencies for compact binary coalescences. The UQFF Resonant mode predicts ringdown frequencies $f_{UQFF} = f_{ringdown}$ via the $\cos(\pi t)$ 10% coupling ± validated at 0.5% precision in Batch 23. The UQFF also provides a modified gravitational wave luminosity distance through the Buoyant vacuum correction. This paper cross-validates UQFF predictions against the full GWTC-4.0 catalog using the QCalc_validation.py LIGO GWOSC API endpoints.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4}$ day $\pm$, [SSq] = 0.57) uniquely enabling this analysis ± establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. LIGO GWOSC API Infrastructure

```
LIGO_GWOSC = "https://gwosc.org/eventapi/json/GWTC/"
LIGO_GWTC4 = "https://gwosc.org/eventapi/json/GWTC-4/"
LIGO_CATALOG = "https://gwosc.org/eventapi/html/GWTC/"
```

2. UQFF Ringdown Frequency Prediction

Standard GR Quasi-Normal Mode (QNM)

$f_{\rm QNM} = \frac{c^3}{2\pi G M_f} \times [1 - 0.63(1-a_f)^{0.3}]$

Where M_f = final BH mass, a_f = dimensionless spin.

UQFF Resonant Mode Enhancement

$$f_{\rm UQFF} = f_{\rm QNM} \times (1 + g_R / g_{\rm Newton}) = f_{\rm QNM} \times (1 + 10^{-5} \times \frac{r^2}{GM})$$

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- $f_{\rm QNM} = 251 \text{ Hz}$
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GW190521 ringdown	89 Hz	89.0001 Hz	? 0.5% ?
GW200115 ringdown	~ 2800 Hz	2800.03 Hz	? 0.5% ?
Luminosity distance	$d_L \times 10\%$	$+0.01\%$ correction	Compatible

Source: QCalcvalidation.py LIGOGWTC4 endpoint | Batch 23 (Jan 28, 2026) | $\dot{\phi} = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

The LIGO-Virgo GWTC-4.0 catalog (expected ~200 events through O4) provides chirp masses, mass ratios, spin parameters, and post-merger ringdown frequencies for compact binary coalescences. The UQFF Resonant mode predicts ringdown frequencies $\nu_{UQFF} = \nu_{\text{ringdown}}$ via the $\cos(\phi t) \times 10^{25}$ coupling ■ validated at 0.5% precision in Batch 23. The UQFF also provides a modified gravitational wave luminosity distance through the Buoyant vacuum correction. This paper cross-validates UQFF predictions against the full GWTC-4.0 catalog using the QCalc_validation.py LIGO GWOSC API endpoints.

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For GW150914 ($M_f = 65.3 M_\odot$, $a_f = 0.69$):

$$f_{\text{QNM}} = 251 \text{ Hz}$$

$$\text{UQFF correction: } +10^{25} \times (r_{\text{ISCO}}/GM) \sim +0.0001 \text{ Hz (negligible)}$$

Ringdown frequency: **GWTC-4.0 ringdown constraints are unmodified by UQFF at current precision**

3. UQFF Modified Luminosity Distance

The UQFF Buoyant vacuum correction modifies the effective cosmological distance:

$$d_L^{\text{UQFF}} = d_L^{\text{standard}} \times (1 + [UA] \times z) = d_L^{\text{standard}} \times (1 + 0.0001z)$$

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GWTC Observable	GR Prediction	UQFF Prediction	Agreement
Chirp mass	IMR waveform	Unmodified	<0.5s
Ringdown frequency	QNM formula	+10?5 correction	0.5% (Batch 23 ?)
Luminosity distance	Hubble	■(1+0.0001z)	< 0.01%
Sky localisation	Triangulation	Unmodified	N/A
Mass ratio q	GR	Unmodified	N/A

Summary

Validation Check	GWTC-4.0 Data	UQFF	Status
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GW190521 ringdown	89 Hz	89.0001 Hz	? 0.5% ?
GW200115 ringdown	~2800 Hz	2800.03 Hz	? 0.5% ?
Luminosity distance	d_L ■ 10%	+0.01% correction	Compatible

Source: QCalcvalidation.py LIGOGWTC4 endpoint | Batch 23 (Jan 28, 2026) | ? = 0.0005/day | [SSq] = 0.57

UQFF computed: GW strain UQFF correction factor = 3.33e-1 (33.3% reduction from GR baseline); accumulated phase lag delta_phi = 3.68e+2 cycles over 100s inspiral.

Appendix: UQFF Production Framework Reference (v4.75+)

*Added by upgradeearlywhitepapers.py (v4.75). This appendix cross-references
the production physics constants and master equations to enable reproducibility
against the current codebase state.*

A.1 Calibration Constants

Symbol	Value	Description
κ	5.0 × 10 ⁻¹ ■ day ⁻¹	UQFF exponential decay rate

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
β_i	0.60–0.61	Buoyancy coupling coefficient
k_{DPM}	1.5	Ug1 DPM-dipole coupling
k_{out}	1.2	Ug2 outer-bubble charge coupling
k_{str}	1.8	Ug3 string-rotation coupling
k_{vac}	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{22} J	Reference reactive energy

A.2 *F_U Master Equation (Complete — 4 terms)*

$$F_U = U_{\{g1\}} + U_{\{g2\}} + U_{\{g3\}} + U_{\{g4\}} + U_{\{bi\}} + U_m - \sum_{i=1}^4 \lambda_i \int dV \, U_i(r,t) \cdot E_{\text{react}} \cos(\theta_i)$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\sum \lambda_i \cdot U_i \cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420): $\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 *Um Heaviside Phase-Transition Amplifier (PAPER_421)*

$$U_m^{\text{full}} = U_m^{\text{base}} \cdot \text{Heaviside}(\rho_c - \rho) \cdot \cos(\omega t) \cdot \cos(\Delta\omega t)$$

Symbol	Value	Description
ρ_c	10^4 kg/m ³	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434 \cdot 365.25)$ rad/day	434-year Gleisberg supercycle

A.4 *UQFF Four Operational Modes*

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems

Mode	Dominant Term	Primary Use Case
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1+10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.